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POSSIBLE IMPLICATIONS OF THE PALAEOMAGNETIC STUDY OF LATE PALAEOZOIC IGNEOUS ROCKS OF NORTHWESTERN CORSICA[1]

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SUMMARY

Palaeomagnetic observations in more detail than heretofore require modification in detail of earlier, tentative, conclusions. The declination is significantly different from that obtained in the Esterel upon rocks believed to be of the same age. The simplest interpretation, a relative rotation of 25°, is consistent with known geology of both regions, the geophysics of the intervening region and with the matching of coast lines. The inclination value, however, is not in accord with the value predicted upon the basis of work in northern Europe.

INTRODUCTION

The paleomagnetic measurements carried out on a pilot suite of Permian "rhyolites" collected during the summer of 1961 by T. P. Ashworth indicated apparent anomalous directions of magnetization. These results, reported by ASHWORTH and NAIRN (1965) seemed to fall into the generally emerging pattern of directions of magnetization inconsistent with the directions recorded from coeval Permian rocks in northern Europe, the Meso-European region of Stille adopted by RUTTEN (1963). Unlike many of the other recorded anomalies there appeared to be no anomaly insofar as inclination was concerned. The interpretation in terms of a rotation of presumed tectonic origin of 53° ± 14° was necessarily tentative in view of the small sample size and the not inconsiderable scatter remaining after demagnetization.

As DUBOURDIEU (1962) had independently drawn attention to faunal affinities between the Lower Palaeozoic assemblages of Sardinia and the Montagne Noire region of southern France (and favoured a displacement), and since rotation of

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Corsica follows directly from the application of Carey's orocline hypothesis (CAREY, 1955), a follow-up study seemed necessary to establish on a sounder quantitative basis whether the palaeomagnetic evidence was indeed consistent with rotation and with the magnitude implied, and to examine the implications of rotation geologically and geophysically. During the summer of 1964, a detailed collection was made of not only the rhyolitic extrusives, but also of some extrusive andesites and both acidic and basic intrusives. The initial results from a further rock body, the Ota granodiorite, and preliminary measurements on some of the rhyolites have already been reported (NAIRN and WESTPHAL, 1967 a, b). The work was made possible by a NATO research grant which is here gratefully acknowledged. In the following year a lithologically similar succession of Permo-Carboniferous beds in the Esterel was examined under the guidance of Dr. J. D. A. Zijderveld who had already completed detailed palaeomagnetic and geological work on the Permo-Carboniferous of that region.

The present paper is divided into two sections. The first, purely palaeomagnetic, reports the results of the investigation of the suite of samples collected in 1965, whilst in the second section pertinent geological and geophysical data relevant to the interpretation of the palaeomagnetic results is reviewed.

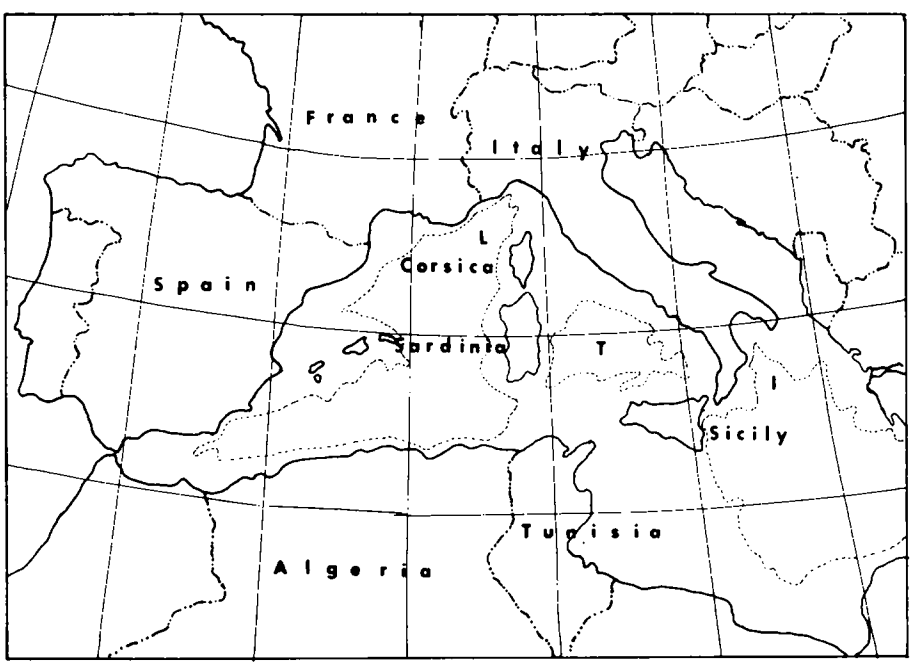


Fig.1. Sketch map of the western Mediterranean and surrounding countries. The 6,000-ft. isobath is drawn. *L* = Ligurian Sea; *T* = Tyrrhenian Sea; *I* = Ionian Sea.

Part I: GENERAL GEOLOGY

Corsica is a mountainous island in the Gulf of Genoa. It lies in the angle between the French and Italian coasts, and with the island of Sardinia immediately to the south, separates the Tyrrhenian Basin from the rest of the western Mediterranean. The islands lie on a submerged platform which connects them to the north and east with the Italian mainland. The continental shelf, everywhere narrow, plunges steeply off the west coast to depths of 2,500–2,700 m in the sea area, the Ligurian Sea, separating Corsica from the coast of Provence (see Fig.1).

For descriptive purposes a threefold division of the main geological features of Corsica is recognized by most geologists since the time of HOLLANDE (1877). As will appear, the significance and the geological interpretation of these units is not so generally accepted. The units are as follows:

(*1*) *The Hercynian granitic massif.* This forms the greater part of the island, roughly everything south and west of a line from Ile Rousse to Solenzara. Over most of this area, because of a dense cover of maquis and "eboulis", save in road cuttings and along the coast, outcrops are rare. The upper age limit of the granite is set by the occurrence of overlying Permo-Carboniferous sediments and volcanics in the region north of Porto. Here the oldest sediments are top Westphalian or Stephanian in age, but near Argentella some 16 km north-northeast of Osani, Dinantian and probably top Devonian rocks are known (KRYLATOV, 1963; KRYLATOV and MAMET, 1966). The granite metamorphoses rocks to which an Ordovician or Silurian age has been attributed (BODENHAUSEN, 1955; KRYLATOV, 1963). The occurrence of gabbroic and dioritic rocks within the massif are obvious indications that it is not as simple geologically as the terminology implies. According to LAPADU-HARGUES and MAISONNEUVE (1963), the eruptives of Monte Cinto, the Permian rhyolites, may be related to the igneous events which earlier produced certain microgranites, thereby suggesting a Dinantian or even younger age for the microgranites.

(*2*) *The "dépressions intermédiaires"*, a series of basins or depressions with a Mesozoic or Cenozoic fill, occur along a line of separation between the granite massif and the schistes lustrés. In some basins Late Cretaceous has been recorded, but Oligocene and Miocene rocks are more common.

(*3*) *The "schistes lustrés"* form the northeastern portion of the island. They contain ophiolite, greenstone and diabase masses. The age of the schists is unknown, and they have been regarded as Hercynian or older by some, and as Tertiary by others; DENIZOT (1958), in the discussion of BENEO's (1954) paper, states their age to be Triassic. BENEO (1954) considered similar schistes lustrés on Elba to be "ante-Hercynian" and to be covered by an ophiolitic "Alpine" nappe. The number and magnitude of nappes and thrusts has also varied according to author (see HOLLANDE, 1877; BROUWER, 1950, 1957; DENIZOT, 1958; LAPADU-HARGUES and MAISONNEUVE, 1963).

THE MONTE CINTO MASSIF

The region lying to the north of Porto, where the outcrops of younger Palaeozoic sediments cover an area of some 400 km^2 is, fortunately, better known than many regions in Corsica. Here two main granite types can be recognized, a biotite granite and a rose-colored variety which form the picturesque relief between Porto and the Calanches de Piana. In the region of the Permian outcrops occurs a third and younger microgranite, the fine-grained varieties of which often closely resemble the rhyolites.

The Permo-Carboniferous succession is best developed in the region of Osani, although outcrops are poor. Here in the sandstones and occasional shale, a coal horizon was at one time locally exploited. The floral evidence from this coal is the sole dated stratigraphic horizon within the succession and suggests a Late Westphalian or Stephanian age. Above the predominantly arenaceous beds occur tuffs and andesitic lavas. The lavas become important towards the top of the succession. In all, five flows were found in the region immediately southwest of Osani; as outcrops are sporadic, further flows may and probably do occur. In thin section the lavas show flow structure with plagioclase and pyroxene phenocrysts, the latter generally altered.

The contact between these beds and the overlying rhyolites is not exposed, but the concordance of dips near Osani suggests a conformable succession. On the Scandola peninsula, however, the rhyolites overstep the sediments and come to rest with discordance upon the granite or older sediments. The rhyolites are thus certainly post Lower Stephanian, most probably Permian, although in the absence of radiometric dates their exact age remains uncertain.

The nature of the rhyolite succession has been debated, DEPRAT (1905, 1906–1907) considering some horizons as conglomerates of volcanic material. Subsequent work, particularly that of BODENHAUSEN (1955) shows that in addition to agglomerates, there are welded tuffs (ignimbrites) and true lavas. The flows (ignimbrites) seen on the Osani and Scandola peninsulas show a close resemblance with the main R3 "rhyolite" of the Esterel which reaches its maximum development in the eastern Esterel and is well exposed along the coast west of Cannes (cf. BORDET, 1951). The location of the vents from which the material was emitted are unknown. DEPRAT (1906–1907) suggested as one possibility the region of Paglia Orba 20 km east of Osani, where dykes become particularly numerous.

The latest phase of activity appears to be that of dyke intrusion, however in the opinion of J. W. A. Bodenhausen (personal communication, 1967) there are probably several generations of acid dykes, the oldest of which may predate the andesitic tuffs of Osani. They are present in large numbers, cutting all the other rocks, and because of their greater resistance to weathering, in the granites they occasionally form remarkable outstanding features. They may locally be so numerous that it is difficult to discern the country rock. They are also variable in

color and structure, sometimes displaying transverse columnar jointing, on other occasions showing spherulitic structures similar to those found in the pyromerides of the Esterel. Multiple intrusions are also common, and in some cases, the same channel has been used by subsequent diabase intrusions. In thickness they may range from 10 m down to a few centimeters.

The diabase dykes, the last intrusive phase, are invariably thin and discontinuous. Their age is unknown. There is a tendency discernible on the foreshore of the Gulf of Girolata near Tuara for those oriented about 330° to be thicker (?tension) than the thinner but longer dykes oriented northeast-southwest. It is also possible by following the dyke pattern to trace a pattern of small-scale displacements, in which the sense of displacement is of a relative movement towards the southeast of the west side of minor faults which here are approximately north south (see Fig.2B).

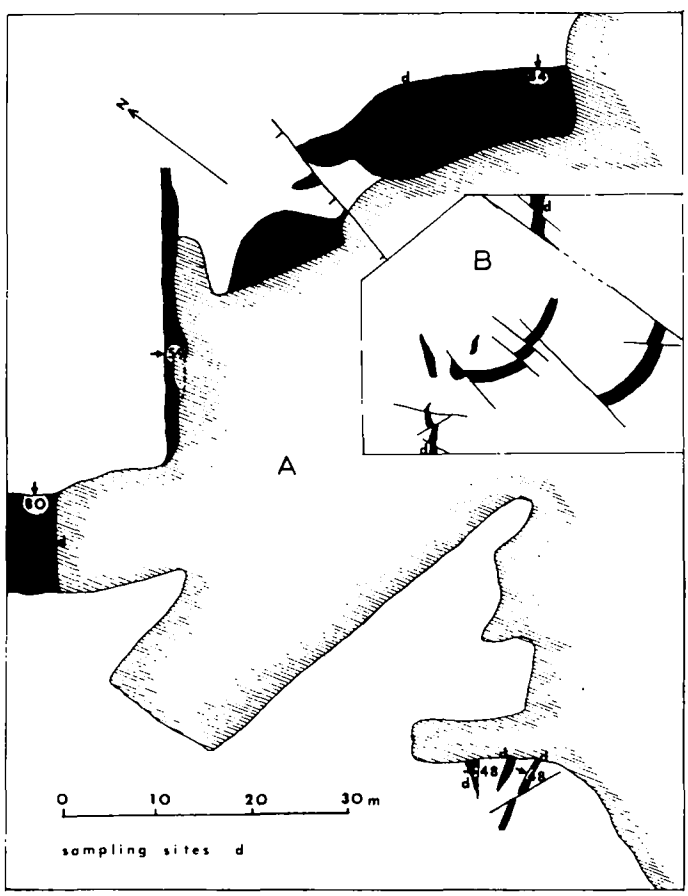


Fig.2. Sketch maps of parts of the Tuara foreshore to indicate the pattern of displacement of the diabase dykes by minor faults. Rhyolite dykes are not drawn. Sampling sites indicated.

There is, according to VAN TELLINGEN (1955), a major fault running west-southwest–east-northeast separating the Porto–Ota zone from the Osani–Monte Cinto region. The fault pattern in the Scandola peninsula with a strike of about 60–70° (BODENHAUSEN, 1955) is similar and parallels the strike direction of the majority of the dykes. A complementary fault set has a north–northwest strike and this is seen again in the microfault pattern on the Girolata foreshore near Tuara.

PALAEOMAGNETIC OBSERVATIONS

Oriented samples, up to seven per site, were collected from 71 sites (Fig.3). The sites were dispersed along route N199 between km 95.0 and 114.4 which gave an excellent suite of dykes, and along footpaths southwest from Osani village where andesites were collected. On the northwest side of the Osani peninsula the welded tuffs were sampled, and on the foreshore near Tuara diabase samples were obtained (Fig.2). From the shore south of Monte Senino further dyke samples

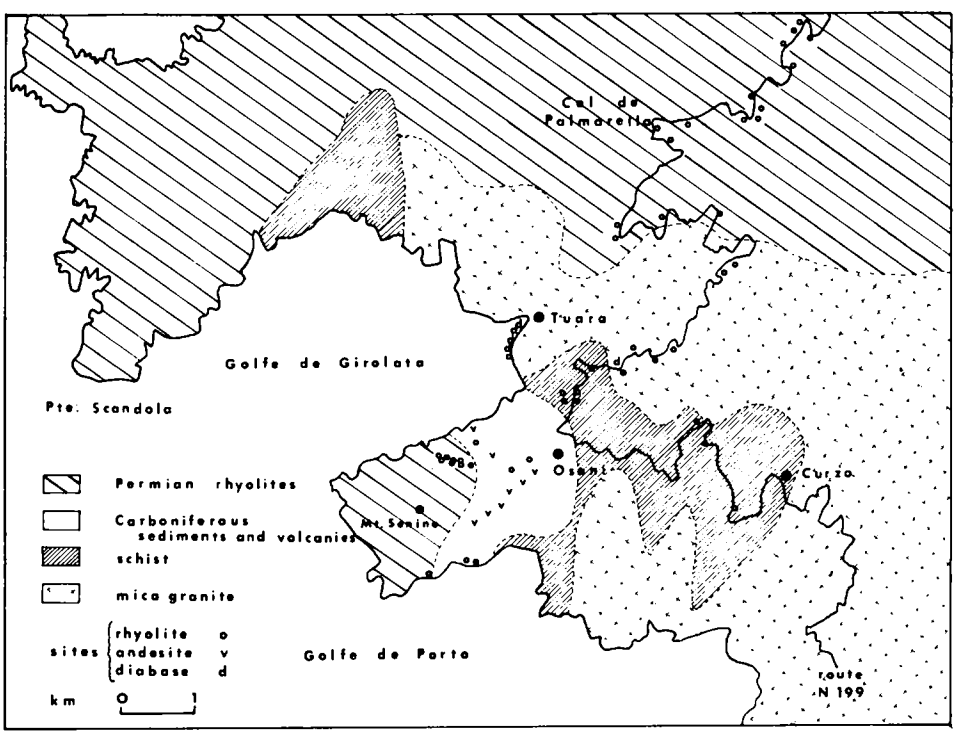


Fig.3. Geological sketch map of the Osani and Scandola peninsulas showing the distribution of the sampling sites.

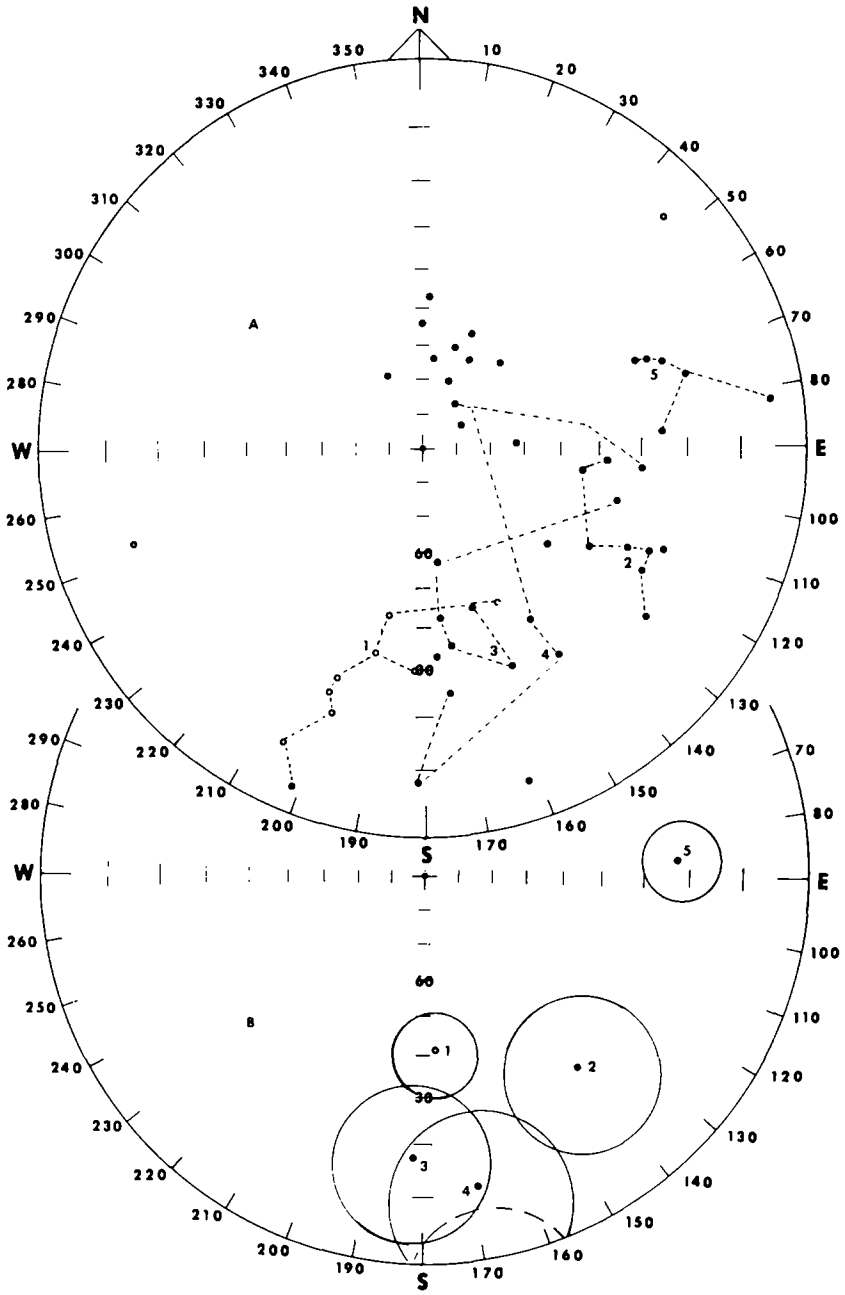


Fig.4. The directions of magnetization of andesite lava samples:

A. N.R.M. of all measured samples.

B. Mean flow direction and circles of confidence after demagnetization in alternating fields.

Some indication of the scatter of N.R.M. directions is given by linking together all samples from each flow whose corresponding mean direction and circle of confidence is given in Fig.4B. The points not connected in 4A responded very little to A.C. demagnetization. By convention plots are on a Wulfnet, open circles represent projection on the Upper Hemisphere, points on the Lower Hemisphere.

were obtained. The results from the gabbrodiorite samples collected near Ota have already been reported (NAIRN and WESTPHAL, 1967a). Precise location of sites was possible by the use of aerial photos in conjunction with hachured maps enlarged to 1/50,000.

Orientation of plane surfaces followed standard geological procedures. From each sample a single long 2.5-cm-diameter core was drilled, and a single core per sample was used for remanence measurements. All remanence measurements were carried out using astatic magnetometers of either two-magnet or three-magnet type. Samples were magnetically cleaned (THELLIER and RIMBERT, 1955; AS and ZIJDERVELD, 1958), some by progressive demagnetization in alternating fields, others by thermal demagnetization (THELLIER and THELLIER, 1959). Correction of observed plane surfaces to the horizontal or for local dip was carried out stereographically.

Each of the four groups of rocks will be discussed in turn from the oldest, the andesites, to the youngest, the diabase dikes.

The andesites

Samples were collected from eight sites, a series of six southwest of the village of Osani, and two sites on the north side of the peninsula. At almost all sites no direct measurement of dip and strike were possible, and for the purpose of applying dip corrections an average value (28° at 243°) based on a score of measurements was used. Although the directions of natural remanent magnetization (N.R.M.) for some sites are reasonably well grouped, in others they are quite scattered (Fig.4A) and there are important differences between the sites. The three well-grouped sites give directions which approximate to the geomagnetic field. The intensity of N.R.M. ranged between 10^{-3} and 10^{-6} e.m.u./cm^3.

In the case of those sites where the directions of N.R.M. were close to the geomagnetic field, demagnetization in alternating field had very little effect and up to the maximum fields employed (800 Oe), the mean directions changed but little. Those sites in which the N.R.M. measurements showed a greater scatter responded to much lower fields, although as the confidence limits in Fig.4B indicate the scatter was still appreciable. Demagnetization curves are figured (Fig.8D). Samples thermally demagnetized up to 500 °C (Fig.8E) gave similar results.

The high dispersion seems most probably due to comparatively hard secondary components of magnetization not removed by 800 Oe fields or by heating to 500 °C. Although local movement of the field samples not easily recognized when outcrops are poor is also a possibility, it is not considered to be the principal cause of dispersion. Changes in the ancient geomagnetic field seem unlikely too, since the sites cover a stratigraphical thickness of no more than 250 m of fairly coarse sediments. It is, however, curious that the flows with the hardest secondary magnetization component are those at the base of the section.

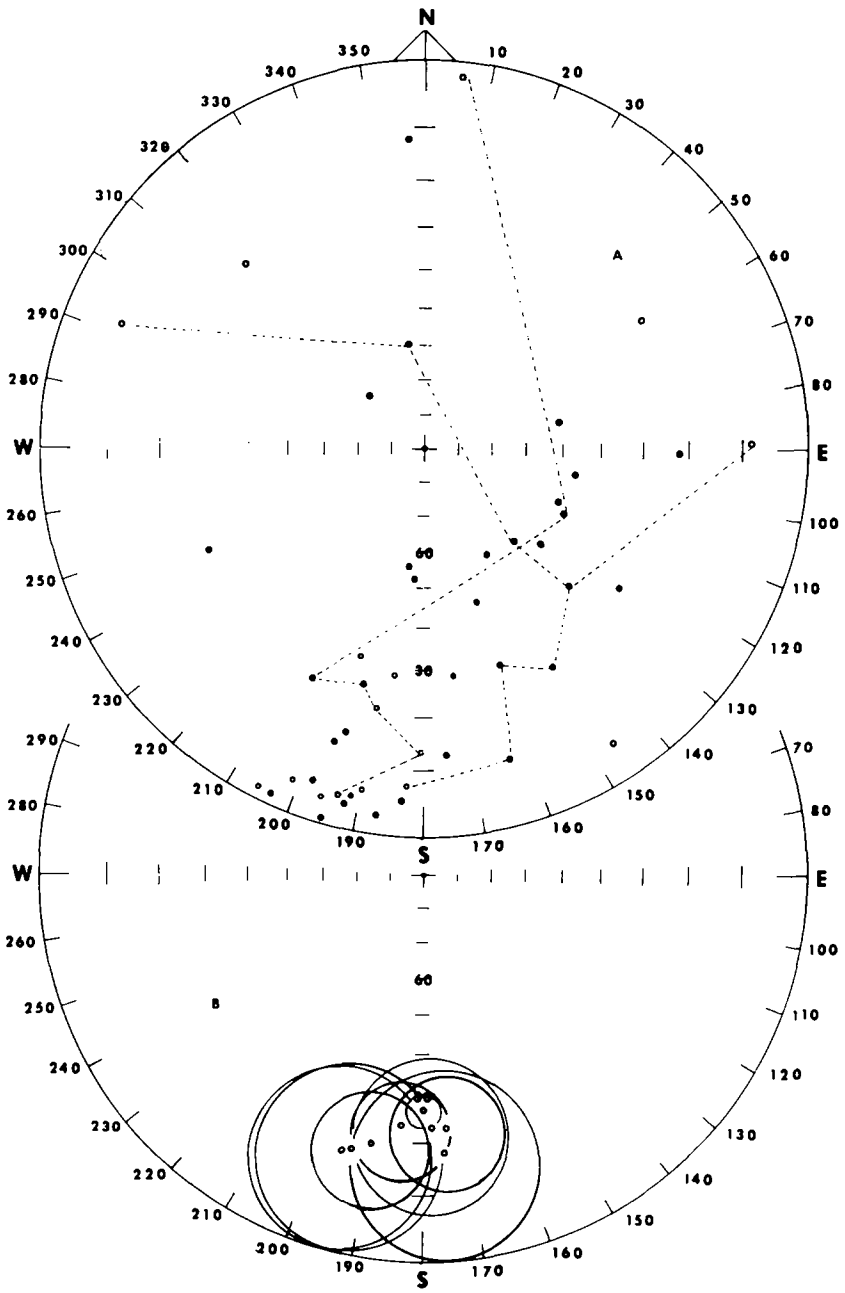


Fig.5. The directions of magnetization of rhyolite extrusives:

A. N.R.M. of all measured samples.

B. Mean flow directions and circles of confidence after demagnetization in alternating fields.

Some indication of the scatter of N.R.M. directions is given by linking together all samples from each of the two flows with the greatest scatter.

The rhyolite extrusives

The ignimbrite flows, were sampled on the north slopes of Monte Senino and in the road cut near the Col de Palmarella. It was, however, sometimes difficult in road cuts to be completely certain that flows rather than thick dykes were being sampled.

The density of N.R.M. was on average about 10^{-5} e.m.u./cm^3 ranging between 3 and $150 \cdot 10^{-6}$ e.m.u./cm^3. The Koenigsberger ratio averaged about 0.5. The dispersion of the N.R.M. direction (Fig.5A) was considerably reduced by A.C. demagnetization in fields of about 250 Oe (Fig.5B). The direction then remained stable in fields up to 700 Oe. Examples of A.C. demagnetization curves are given in Fig.8C. A number of samples were also thermally demagnetized (Fig.8F) with comparable results. A few samples resisted demagnetization up to the limits of the fields applied (700 Oe) with directions which though showing a trend towards the directions displayed by the majority of the demagnetized samples were nevertheless still close to the geomagnetic field directions. These specimens were regarded as possessing a hard secondary magnetization and were not considered further.

The rhyolite dykes

The measurements of N.R.M. (Fig.6A) on the dykes indicated a greater scatter than that present in the rhyolite extrusives. The intensity of magnetization which ranged between $0.1–400 \cdot 10^{-6}$ e.m.u./cm^3 was most commonly on the low side of that range which imposed limits on the demagnetization procedures, for in many cases the intensity of magnetization dropped below measurable limits before completion of demagnetization. The maximum A.C. fields applied ranged from 300 Oe to 1,000 Oe; demagnetization curves are illustrated in Fig.8B.

As in the case of the rhyolite flows, one group of measurements remained close to the present geomagnetic field direction and were not considered further. The majority displayed remanence directions similar to those recorded from the andesitic and rhyolitic flows (Fig.6B). Finally some samples were found in which east southeasterly directions were recorded.

No dip corrections were made, and since the dykes sampled were usually cutting through the granitic massif, none were deemed necessary.

The ferromagnetic minerals present in both the dykes and flows were examined in a limited number of polished sections by Bolfa. Few opaque minerals are to be seen, and these are of small size (20 μ), the greater part of the iron oxide minerals are present in the form of hydrates with an indeterminate amount of titanium. They were presumably derived from the late stage of hydrothermal alteration of magnetite and/or haematite or ferromagnesian minerals, or alteration just subsequent to emplacement while the rock was still cooling. Hydrothermal

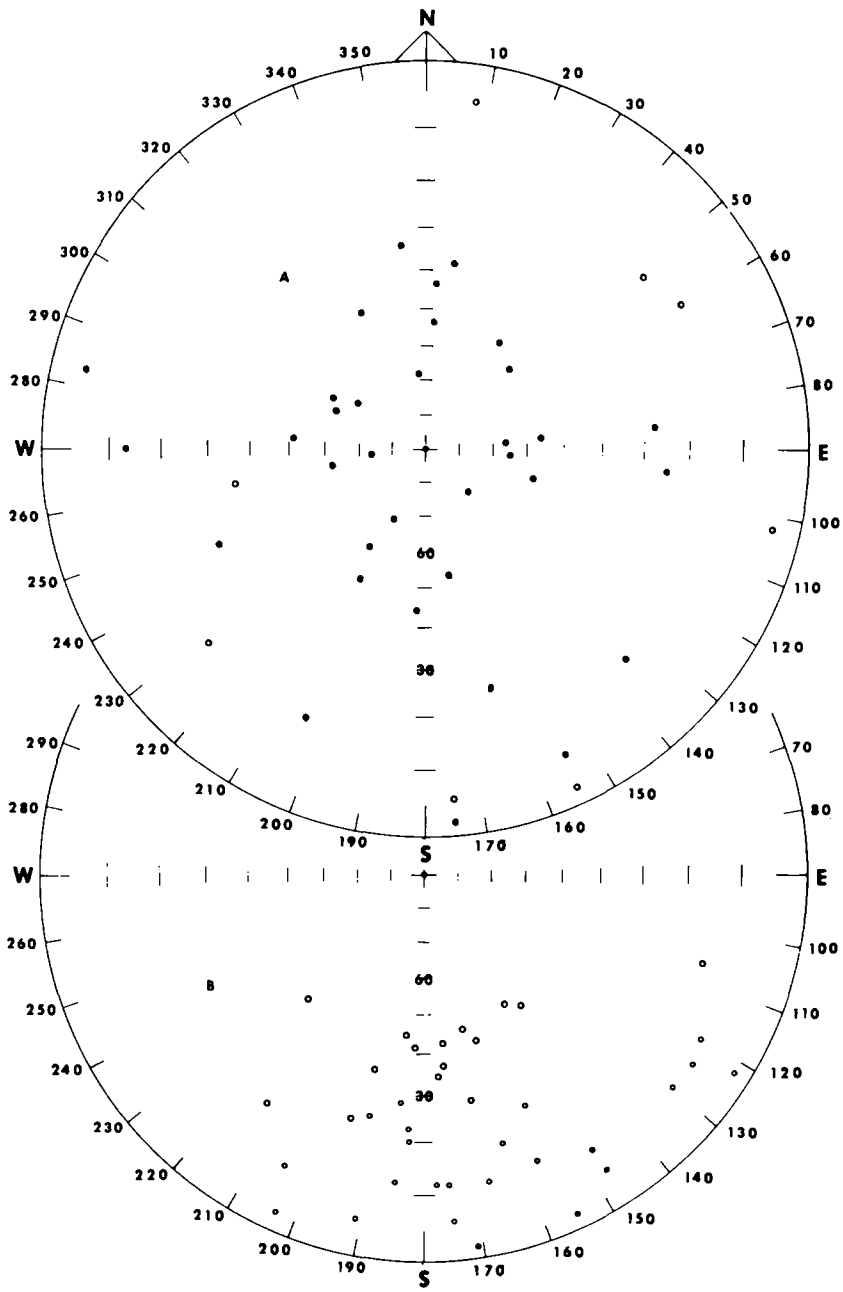


Fig.6. The directions of magnetization of rhyolite intrusives:

A. N.R.M. of all measurements.

B. Directions of magnetization after demagnetization in alternating fields. All samples are shown although in the computation of the mean direction each dyke was given unit weight. As the number of samples per dyke was variable and often small mean directions and confidence circles were not drawn.

alteration also affects other minerals and BODENHAUSEN (1955) describes albitization and even kaolinization of the feldspars.

Such mineralogical changes are consistent with the physico-chemical conditions, high temperature, gas and water vapour content and solid particles from which an ignimbrite formed. It is presumed that the magma from which the dykes were formed must likewise have had a high volatile content. Subsequent alteration, excluding surficial alteration, is possible, but not regarded as significant. The magnetization measured after the removal of secondary magnetic components is therefore more a combination of a C.R.M. (chemoremanent magnetization) and T.R.M. (thermo-remanent magnetization) than solely T.R.M.

During the course of thermal demagnetization there is a steady change in composition as the hydrates lose water and are converted to haematite or magnetite according to the condition of heating. The changes in composition correspond to the observed variations in susceptibility. Samples were heated in air in the earth's field and cooled in field free space, field cancellation being checked prior to each run. Above 500 °C new secondary components appear. These may be the result of the interaction of the earth's magnetic field and the molecular field of the crystal, even though the samples were cooled in field free space.

It is difficult to be specific in the positive identification of the carrier of remanence in every case. J. D. A. Zijderveld (personal communication, 1967) reports the same remanence directions from both hydrate bearing rocks and those in which hydrated iron oxides are absent. This argues not only for the near contemporaneity of the hydration, but also for the presence of magnetite and/or haematite in some rocks. In others the remanence appears to be due in part at least to imperfect spin balance in the hydrate (STRANGWAY et al., 1967, FULLER et al., unpublished reports, University of Pittsburgh).

The diabase dykes

The diabase dykes appear to form the last igneous episode. Like the rhyolite dykes, no close dating time of intrusion is possible. Unlike the rhyolite dykes, they cannot be petrologically related to any other rock type save the Ota gabbrodiorite, a suggestion made by DEPRAT (1905) on rather unsubstantial grounds. These dykes are generally thin and discontinuous, and in combination with their low resistance to weathering when compared with the rhyolites, leads to poor exposure away from the foreshore.

In common with all the other rocks described, the N.R.M. directions (Fig.7A) are scattered, showing that they possess important secondary components of magnetization which may in some cases also be magnetically hard and little affected by A.C. demagnetizing fields. In other cases a field of about 150 Oe is sufficient for cleaning, giving directions (Fig.7B) stable in fields up to about 400 Oe. At higher fields the intensity of the samples was reduced below the level of sensitivity of the magnetometer (Fig.8A).

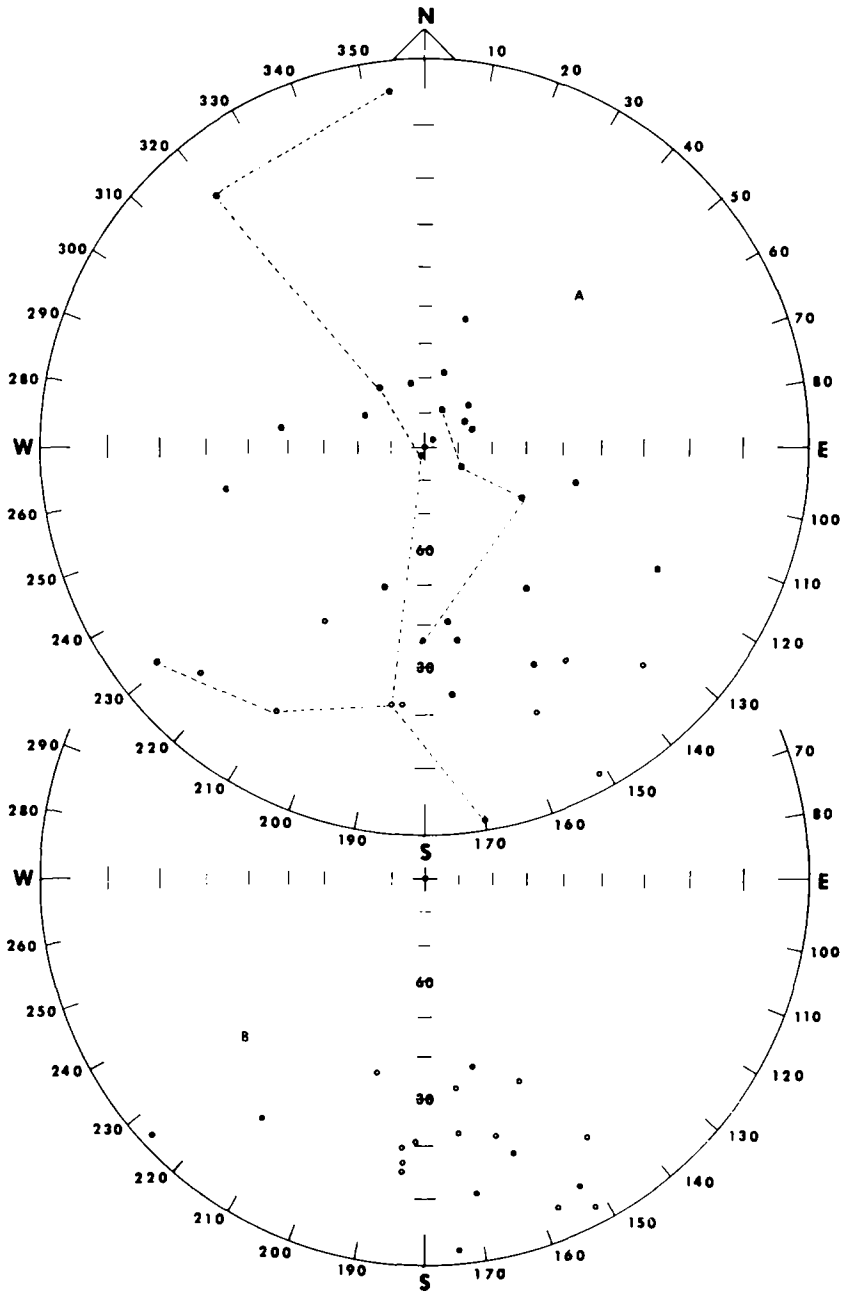


Fig.7. The directions of magnetization of diabase dykes:

A. N.R.M. of all measurements.

B. Directions of magnetization of all samples used in computation after demagnetization in alternating fields. Some indication of the scatter of N.R.M. directions is given by linking all samples from the two dykes with the greatest scatter.

DISCUSSION OF PALAEOMAGNETIC DATA

Directions of magnetization

In the work here reported four groups of directions can be recognized. One has a southerly declination revealed by demagnetization. It is the direction considered to be original and is represented by the data summarized in Table I. The second group comprises rocks whose N.R.M. directions grouped close to the present geomagnetic field and which upon demagnetization proved to have a secondary component sufficiently hard so that it was essentially unaffected by the cleaning techniques employed. This hard component may result from surficial oxidation of minerals like pyrite, or even to thermal remagnetization caused by the burning of the maquis. The third group possessed declinations in the southeast quadrant (120–140°), similar to directions earlier recorded in the rhyolites (ASHWORTH and NAIRN, 1965). These directions were found in the rhyolite dykes and the lowest of the andesite flows and they appeared more clearly after demagnetization. They represent only a small proportion of the rock measured, and their interpretation poses an unsolved problem. If the direction is the resultant of a very hard secondary component comparable in stability with the original magnetization it would not appear to have been acquired in any field comparable to the present, but would require one with a much lower inclination value, of the order of that found in the Triassic, for example. The fourth group of directions are widely dispersed initially and some remain so after demagnetization, so that it may be supposed that some had been affected by lightning remagnetization. Experience has shown that lightning remagnetization effects, although normally susceptible to A.C. cleaning techniques, is not always removed.

Comparisons of directions of magnetization with other results

Ota gabbrodiorite

The directions of magnetization from this rock were reported earlier (NAIRN and WESTPHAL, 1967a). The directions were hard to interpret because of the problem of the age of the magnetization and of the rocks in question. It is now clear that the directions of magnetization are comparable with the Stephanian and probable Permian directions from the acid volcanics. It follows if the views of VAN TELLINGEN (1955) are correct and the rocks are enclaves in the granite, their magnetization is that of the enclosing granite. The granites are certainly older than Stephanian, although LAPADU-HARGUES and MAISONNEUVE (1963) suggest that some of the microgranites may be as young as early Stephanian with the majority of the normal granites probably Dinantian. The regional relationships with the granite massif are so little known that in the absence of radiometric dating any age from Devonian to Stephanian is possible. Although we consider DEPRAT's (1905) conclusions are

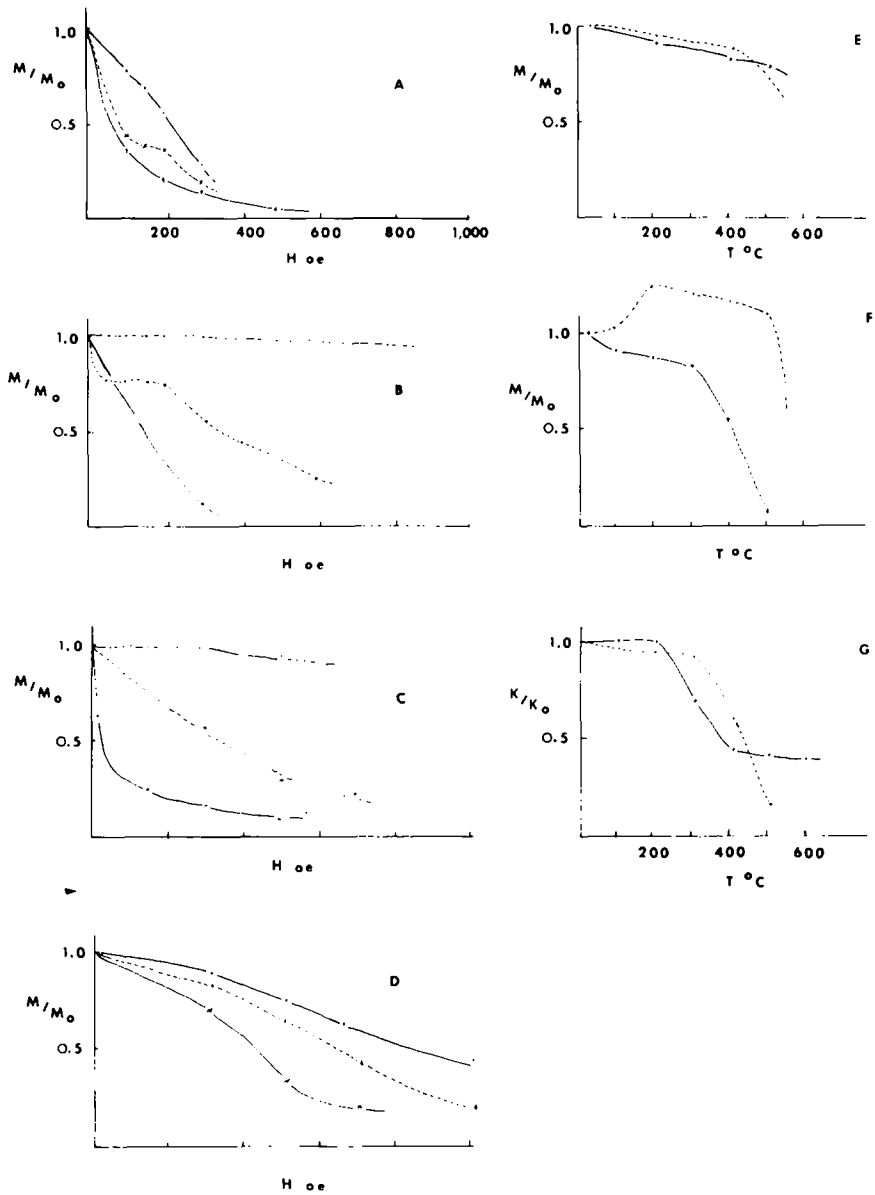


Fig.8. Demagnetization curves of various samples.
A. A.C. demagnetization curves of samples from three sites in diabase dykes.
B. A.C. demagnetization curves of samples from three sites in rhyolite dykes.
C. A.C. demagnetization curves of samples from three sites in rhyolite flows.
D. A.C. demagnetization curves of samples from three sites in andesite flows.
E. Thermal demagnetization curves of samples from two sites in andesite flows.
F. Thermal demagnetization curves of samples from two sites in rhyolite flows.
G. Curve showing the change of susceptibility after each heating of the same samples shown in Fig.8F.

TABLE I

SUMMARY OF DATA FROM CORSICAN AND OTHER ROCKS

Rock unit	*No. of sites*	*Mean*		α_{95}	*K*	*R*	*Virtual pole*		*dp*	*dm*
		declination	*inclination*							
(1) diabase dykes	9	183	− 7	20	7.4	9.924	51 °N	176 °W	10	20
(2) rhyolite dykes	16	177	−19.5	10	15	14.996	57.5 °N	166 °W	5.6	10.6
(3) rhyolite extrusives	8	183.6	−21.6	6.1	83.2	7.916	58 °N	179 °W	3.5	6.4
(4) andesite extrusives	4	166.6	+ 4.0	42.6	5.6	3.461	54 °N	149 °W	21	43
(5) Ota gabbrodiorite	1*	189.2	−13.1	—	—	—	53.5 °N	173 °E	—	—
(6) Osani peninsula	15**	141	± 8.5	14	8	—	31.5 °N	124 °W	7	14
(7) Esterel rocks	—***	207	−22	3.5	—	—	51 °N	129 °W	2	37
Mean of 1–4		174	−10	24	16	3.812	—	—	—	—
Mean of all sites in 1–4		177	−14	8.4	8.6	33.688	—	—	—	—

* Treated as a single unit (NAIRN and WESTPHAL, 1967a).
** Results from ASHWORTH and NAIRN (1965).
*** Data from J. D. A. Zijderveld (personal communication).

incorrect on geological grounds, the palaeomagnetic evidence would not exclude his correlation of the gabbrodiorite with the diabase dykes of the Tuara foreshore.

It therefore appears in the light of the present work, that following VAN TELLINGEN'S (1955) thesis, the local geomagnetic field at the time of granite emplacement was essentially the same as that during the Permo-Carboniferous, which may be construed as favouring a low Carboniferous intrusion date rather than some time in the Devonian. The palaeomagnetic data is also consistent with DEPRAT'S (1905) conclusions of a Late Palaeozoic date of intrusion of the Ota gabbrodiorite. Whilst on the basis of published work Van Tellingen's conclusions appear to have the sounder geological basis, final resolution would seem to demand radiometric and further field studies.

The Esterel

The closest rocks both in terms of distance and age to the Corsican Permo-Carboniferous outcrop are in the Esterel. Since the early studies of RUTTEN et al. (1957) and ROCHE (1957), consideration of the volcanic rocks and their stratigraphical relationships have formed part of a detailed study by J. D. A. Zijderveld (personal communication, 1967). Zijderveld's results based on close stratigraphical control, detailed sampling and careful examination provide an excellent basis for comparison. This comparison (Table I) shows that there is a significant difference in mean declination between the two regions, although the inclination values are almost identical. Neither the result from the Esterel, nor the present results from Corsica, are consistent with the pattern of Permian inclination and declinations from Stille's meso-Europe region (RUTTEN, 1963). According to the palaeo-isoclinic map of VAN HILTEN (1962) an inclination value of +6° should be anticipated.

If we consider first the declination differences between the Esterel and Corsica, three possible explanations come to mind.

(*1*) A marked age difference between the two groups of rocks.

(*2*) Marked secular variation may affect the mean in rocks regarded as coeval from the geological standpoint.

(*3*) Relative displacement.

It is true that there are severe limitations in the dating of volcanic activity in both areas. The principal information available is palaeobotanical. The base of the section in the Esterel is Stephanian, but according to Zijderveld, there is an important break in the sequence with the rhyolites dating about Middle Permian on the basis of intercalated beds with plant remains. There is less information in Corsica, where on palaeobotanical grounds the coal is placed as Westphalian/Stephanian. Above the coal horizon occur the andesitic tuffs and lavas and then the rhyolites. As no contacts are exposed, there is no evidence one way or the other whether the rhyolites are conformable, or whether there is a break in the succession, as is suggested in the Esterel.

Other measurements in other parts of the world do not indicate any appreciable differences over the time interval involved, and in consequence an age difference does not appear to be the cause of the declination difference.

The time range represented by the samples collected in both areas can be confidently regarded to adequately cover the period usually considered necessary to average out secular variation, for sites are numerous and widely distributed. A magnetic anomaly as a cause of the difference would require gradients of the order of 8,000γ/100 km (80γ/km) in a magnetic field of present day intensity, a figure well in excess of anything known at the present time. Thus there remains the possibility of relative displacement. The simplest way of satisfying the requirement involves a relative rotation between the two regions as was suggested by the early study (ASHWORTH and NAIRN, 1965), although the magnitude is only about half of that proposed. In the following section the suggestion of rotation will be examined independently.

Part II: DISCUSSION OF ROTATION

Working upon the assumption that rotation may be the explanation of the differences in the declination values recorded between the Esterel and Corsica, the geological and geophysical consequences are now reviewed.

Coastline matching

The work of BULLARD et al. (1965) spectacularly demonstrates the matching of the coastlines of South America and South Africa. Using a modification of that programme, a computer matching of the Esterel and Corsican coastlines at the 1,000 m isobath with 78 control points on each profile was carried out on the Cambridge computer by Dr. A. C. Smith. Two results are given below, one based upon the originally selected rotation point, the second based upon a modified rotation point and the elimination of four non-overlapping end points on the Esterel margin with the interpolation of four additional points:

Initial rotation point 44.8 °N 10.4 °E rotation 33.51 ° misfit 4.44 °
Modified rotation point 44.7 °N 10.9 °E rotation 30.87 ° misfit 1.98 °

Although the matching is not perfect (an angular misfit of 2 ° represents a physical distance of over 50 km at the distant end of the profile), there is a remarkable agreement between the rotation values obtained from the computer match of the coastlines (30.87 °) and the difference between the mean paleomagnetic values of the Esterel (Zijderveld) and the mean Corsican value (30 °). Fig.11 represents the cartographic fit of the coastlines. In preparing this figure it was

found that the chief effect of varying the isobath used in matching was to alter the separation. It made little difference to the overall result. The difference between the cartographic fit illustrated and the computer fit is small, and resides in the fact that while in the former, overlap is minimized and the separation is about the same at both ends of the profile, in the latter the angular misfit is minimized.

Geological matching

Some indication of the matching of the geology is shown in Fig.11. The most striking feature to emerge from the coastline fit is the apparent continuity of the Permo-Carboniferous outcrops which results even to the extent of the apparent alignment of the faults bounding the Permian in both areas. Detailed comparison is handicapped, however, by the lack of detailed descriptions of the geology of Corsica. The most complete Permo-Carboniferous in the two areas concerned is in the western Esterel. The sedimentary sequence becomes attenuated eastwards, and if the reconstruction is to be believed, in Corsica is finally cut out by the overstep of the younger extrusives on to older rocks. In the Esterel the rhyolites become of greater and greater significance from west to east, and along the coast near Cannes the massif R3 horizon crops out. In Corsica, lithologically similar massif rhyolites form the Scandola and Osani peninsulas. In general the same lithological varieties in the acid extrusives is found in both areas. Thus, the general pattern of facies change along the strike in the Esterel is perfectly consistent with the observed geology in Corsica. By the same token however, direct close correlation is impossible and the evidence is therefore circumstantial.

When attempting to examine the geological consequences in still older rocks, the almost total lack of data from a large part of the Corsican Hercynian granite massif is a severe handicap. In Provence there is, passing from Toulon in the west eastwards to the Maures massif which is separated by the Argens depression from the Esterel and the Tanneron massif in the Esterel, a gradual increase in the grade of metamorphism from phyllites to migmatites and granites. The Hercynian massif of Corsica could be interpreted as the final stage in the classical metamorphic sequence, as a completely remobilized granite. The separation is however greater than is the case with the Permo-Carboniferous and the evidence is even more circumstantial.

Crustal structure

In both the Esterel and Corsica the coastal margins are strongly faulted. Between the two lies the Ligurian Sea with a relatively constant depth of 2,500–2,700 m. Seismic refraction studies (MURAOUR et al., 1966) suggest the faulting of the western and northern coastal regions of Corsica appears to be step faulting with vertical displacements of the granitic margin of several hundred metres. The

submarine topography indicates this may be true along the whole of the west coast. The crust under Corsica is of continental type and is continuous with the Italian mainland being linked to it by a submerged ridge from which rises the island of Elba. Gravimetric studies (CORON, 1954) show minimal residual isostatic anomalies for a compensation depth of 30 km.

The primary massif in Provence disappears northwards under, or is cut off by, the thrust zone of the Alps and of Provence. Most of the epicentres recorded appear to be associated with this zone (Fig.9). The faulted Argens depression separates this massif into the Esterel in the east and the Maures massif in the west.

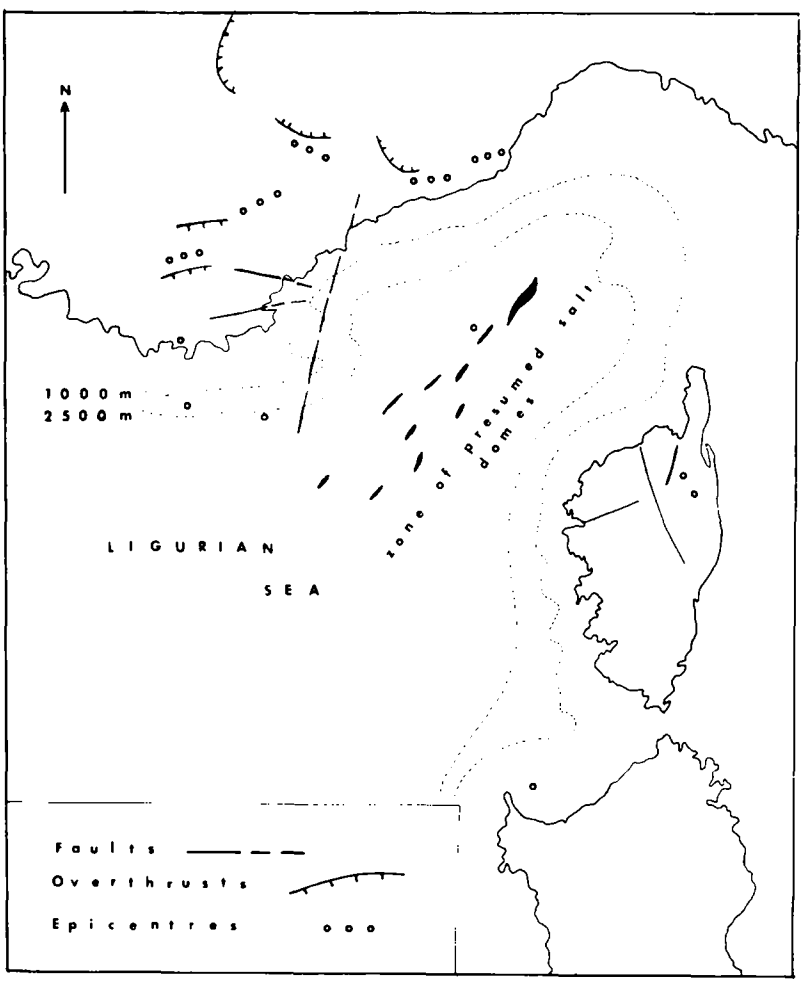


Fig.9. Sketch map of the Provence coast and Corsica. Some of the principal Alpine thrust zones and faults are indicated together with earthquake epicentres. Isobaths for 1,000 m and 2,500 m are drawn. The zone of presumed salt domes along the axial region of the Ligurian Sea is also shown.

The basement of the Esterel as exposed in the Tanneron massif is composed of gneiss and mica-schist, similar to that exposed in the Maures massif (GUEIRARD, 1962). The difference then is in the sedimentary–volcanic sequence (BORDET, 1951, 1966) of the Esterel. RECQ (1967) has recently calculated a crustal thickness of 30 km under Provence with the Moho gradually dipping northwards. This agrees with the interpretation of isostatic anomalies.

Seismic study of Lg surface wave, between Europe and Africa indicates an oceanic structure for the floor of the Western Mediterranean Sea (PETERSCHMITT, 1967). Detail of the structure of the floor of the intervening Ligurian Sea has been obtained from the study of seismic reflection and refraction profiles (FAHLQUIST, 1963). The Moho is placed at a depth of 11 km overlain by a 4 km crust of probable basaltic composition and a comparable thickness of sediments. The depth of the Moho is in reasonable agreement with that predicted from the study of the dispersion curves of surface waves (PAYO, 1966).

The seismic reflection studies indicate the existence of domes piercing the sedimentary rocks in the central part of the Ligurian Sea. The domes are oriented northeast southwest paralleling the faults in the region, and may be of salt (ALINAT et al., 1966), since they are not associated with any marked magnetic anomaly although narrow basaltic intrusions are not ruled out. The magnetic anomalies which have been recorded (:: 250 γ) tend to occur in the marginal regions and an obvious interpretation is that they represent the intrusion of basic material in the coastal fault zone.

The general conclusion which is to be drawn from the preceding review of geological and geophysical data pertinent to the idea of rotation is therefore inconclusive. From the geological standpoint rotation may not produce any anomalies, and if it does produce a reasonable matching, the facies changes preclude definite correlations. Geophysically the floor of the Ligurian Sea is oceanic in accord with the failure to recognize the "Lg" phase and any other result would be inconsistent with rotation.

INTERPRETATION

There is general agreement between the result of palaeomagnetic measurements from four separate rock groups, andesitic and rhyolitic extrusives, acidic and basic minor intrusives. Within the limits of geological dating, these rocks are Permo-Carboniferous. The mean direction of magnetization is given in Table I. Two results given are based upon the means of the four groups and the other value obtained by giving each site unit weight. There is, however, a danger in placing too much weight on a mean from groups of rocks whose exact time stratigraphic relationships with one another are not well defined.

The occurrence of a small proportion of rocks of which the mean direction of magnetization lies in the south east quadrant poses a problem not yet solved.

It is tentatively suggested that they may be due to a hard secondary component of magnetization possibly acquired in Early Mesozoic times, rather than marked secular variation for which there is, so far, no evidence elsewhere.

There is a significant difference between the mean declination of the Corsican rocks and their coeval equivalents in the Esterel which is considered to be the result of a predominantly rotational movement. The coastal margins of both regions are faulted and the Ligurian Sea is underlain by oceanic crust. The cartographic matching of the coasts involves a rotation of the same order as the rotation implied palaeomagnetically. The rotation is not incompatible with the observed regional geology but the geological matching is not conclusive, although many

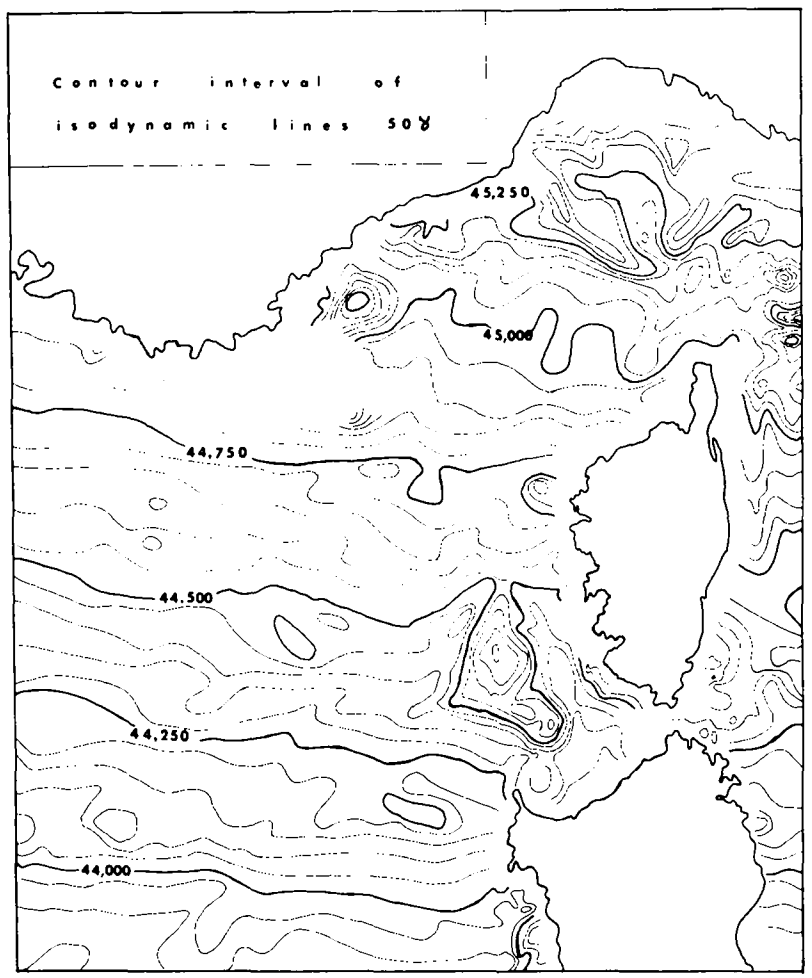


Fig.10. Isodynamic map of the Ligurian Sea area, abstracted from the magnetic maps published by the U.S. Naval Office. The map illustrates the generally smooth pattern found in regions with a water depth in excess of 2,500 m.

similarities are apparent. The only data which requires movement is the palaeomagnetic.

Previously displacement in the same sense as the rotation implied by the palaeomagnetic results had been suggested by DUBOURDIEU (1962) to account for faunal similarities in lower Palaeozoic rocks between the succession in the Montagne Noire and that found in Sardinia which does not seem to the authors a very compelling argument.

Rotation was also a logical consequence of CAREY's (1955) reconstruction of the pre-Alpine Tethys by restoring the six European oroclines.

Assuming the case for rotation, then it is possible to place certain limits upon the date of separation. Pliocene marine rocks are known in western Corsica in the bay of Ajaccio (GLANGEAUD, 1966), and marine Oligocene rocks are reported from Bonifacio. The most important single piece of information is the thickness of the sedimentary pile flooring the Ligurian Sea. In the central region the sediments are pierced by domes which are oriented along northeast–southwest lines. It has been suggested that these domes may be of salt (ALINAT et al., 1966; GLANGEAUD et al., 1966) certainly they are not associated with important magnetic anomalies. Aeromagnetic surveys (Fig.10) show that the most important magnetic anomalies are along the continental margins and may thus be interpreted as intrusion along the faulted coastal margins. Ages for the salt ranging from Triassic to Tertiary have been proposed and GLANGEAUD et al. (1966) have suggested that the salt is Triassic and that these and the marginal intrusions rose along tension faults of his "antéclise ligurienne".

The thickness of the sediments is such that the inception of separation would seem of necessity to be at least of Mesozoic age; even a Late Permian date is not impossible although an estimate of age based on sedimentation rates alone is suspect. Indeed if the "salt" is truly Triassic as GLANGEAUD et al. (1966) affirm, the separation of Corsica must of necessity be at least Late Permian. The relationship of the movement to Alpine tectonics is thus by no means as direct as the orocline concept might suggest.

If in discussing rotation the implicit assumption is that the mean declination value for northern Europe can be used as a standard of comparison some indication of the body being rotated can be obtained. This direction, 194°, leads to the conclusion that both the Maures-Esterel and Corsica were involved in the rotational movement (Fig.11). This interpretation cannot be more than tentative because of the problem of the inclination. The inclination values from Corsica and the Esterel are virtually the same but are inconsistent with calculated value based upon the mean northern European figure. This can be observed on the isocline map of VAN HILTEN (1962), which predicts an inclination of +6° for the present position of Corsica. At the present state of knowledge, while the palaeomagnetic result is consistent with the translational movements in the Tethyan region of the type proposed by DE BOER (1965) and discussed by IRVING (1967), the presence

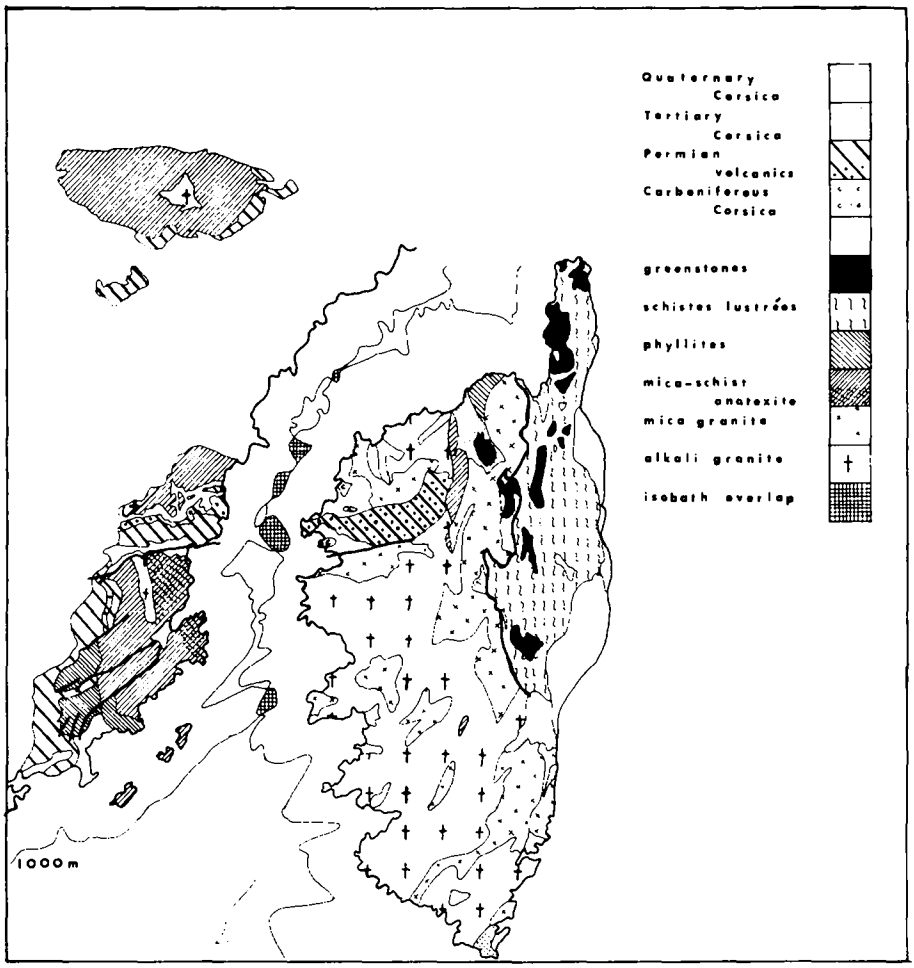


Fig.11. An attempted reassembly of Corsica and the Maures-Esterel based on matching the 1,000 m isobath. It eliminates the anomaly between the observed palaeomagnetic declination. The principal features of the geology of both regions as indicated were taken from maps of the French Geological Survey. Note that in the Esterel although rocks are referred to both the Carboniferous and Permian they are both combined as Permian, while on Corsica they are separated. The unshaded region in Provence has a cover of Mesozoic to Recent rocks and was involved in Alpine tectonic movements.

of the thick sedimentary pile in the Ligurian Sea and the tectonic structures defined seismically impose severe restrictions and make such an interpretation premature.

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REFERENCES

ALINAT, J., GIERMANN, G. et LEENHARDT, O., 1966. Reconnaissance seismique des accidents de terrain en mer Ligure. *Compt. Rend.*, 262B: 1311–1314.

AS, J. A. and ZIJDERVELD, J. D. A., 1958. Magnetic cleaning of rocks in palaeomagnetic research. *Geophys. J.*, 1: 308–319.

ASHWORTH, T. P. and NAIRN, A. E. M., 1965. An anomalous pole from Corsica. *Palaeogeography, Palaeoclimatol., Palaeoecol.*, 1: 119–125.

BENEO, E., 1954. Sull'eta dell'orizzonte degli «schistes lustrés» nella costa orientalé dell'isola d'Elba. *Congr. Géol. Intern., Compt. Rend., 19e, Algiers, 1952*, 13: 63–66.

BODENHAUSEN, J. W. A., 1955. *Géologie de la Région au Nord du Golfe de Porto (Corse)*. Thesis, Univ. Amsterdam, 98 pp.

BORDET, P., 1951. Étude géologique et pétrographique de l'Esterel. *Mém. Carte Géol. France*, 1951: 207 pp.

BORDET, P., 1966. *L'Esterel et le Massif de Tanneron*. Hermann Paris, 114 pp.

BROUWER, H. A., 1950. Sur la tectonique de la Corse. *Koninkl. Ned. Akad. Wetenschap., Proc., Ser. B*, 53: 1–6.

BROUWER, H. A., 1957. Contribution à la géologie de la Corse. *Geol. Mijnbouw*, 19: 317–328.

BULLARD, E., EVERETT, J. E. and SMITH, A. C., 1965. The fit of the continents around the Atlantic, *Phil. Trans. Roy. Soc., London, Ser. A*, 258: 4–51.

CAREY, S. W., 1955. The orocline concept on geotectonics. *Papers Proc. Roy. Soc. Tasmania*, 89: 255–288.

C.N.R.S., 1963. Recherches seismologiques dans les Alpes occidentales au moyen des grandes explosions (en 1956, 1958, 1960). Dans: H. CLOOS et Y. LABROUSTE (Rédacteurs), *Groupe d'Étude des Explosions Alpines—Centre Natl., Rech. Sci.*, 12(2): 241 pp.

CORON, S., 1954. Contribution à l'étude de la pesanteur en France (La Corse). *Sci. Terre*, 2(4): 113–133.

DE BOER, J., 1965. Paleomagnetic indications of megatectonic movement in the Tethys. *J. Geophys. Res.*, 70: 931–944.

DENIZOT, A., 1958. La structure géologique de la Corse. *Études Corses*, 19: 80 pp.

DEPRAT, J., 1905. Étude pétrographique des roches éruptives de Corse. *Bull. Serv. Carte Géol. France*, 114: 39–45.

DEPRAT, J., 1906. Les éruptions carbonifères et permiennes en Corse. *Bull. Soc. Géol. France*, 6: 66–74.

DEPRAT, J., 1906–1907. Étude des roches éruptives carbonifères et permiennes du N–O de la Corse. *Bull. Serv. Carte. Géol. France*, 117: 279–363.

DUBOURDIEU, A., 1962. Le déplacement de l'Europe occidentale. *Compt. Rend.*, 254: 510–512.

FAHLQUIST, D. A., 1963. *Seismic Refraction Measurements in the Western Mediterranean Sea*. Thesis, Mass. Inst. Technol., Cambridge, Mass., unpublished.

GLANGEAUD, L., 1965. Morphotectonique de la terminaison sous-marine orientale des Maures et de l'Esterel. *Compt. Rend.*, 261: 4795–4798.

GLANGEAUD, L., 1966. Sur les grands ensembles structuraux de la Mediterranée occidentale d'après les données de Geomede, I. *Compt. Rend.*, 262d: 2045–2048.

GLANGEAUD, L., SCHLICH, R., PAUTOT, G., BELLAICHE, P., PATRIAT, P. et RONFARD, M., 1965. Morphologie, tectonophysique et évolution géodynamique de la bordure sous-marine

des Maures et de l'Esterel. Relation avec les régions voisines. *Bull. Soc. Géol. France*, 4(7): 998–1009.

GLANGEAUD, L., ALINAT, J., POLVÈCHE, J., GUILLAUME, A. et LEENHARDT, O., 1966. Grands structures de la mer Ligure, leur évolution et leurs relations avec les chaînes continentales. *Bull. Soc. Géol. France*, 8: 921–937.

GUEIRARD, S., 1962. *Le Massif des Maures, Guide Geologique*. Hermann, Paris, 78 pp.

HOLLANDE, M., 1877. Géologie de la Corse. *Ann. Sci. Géol. (Lille)*, 9(2): 114 pp.

IRVING, E., 1967. Paleomagnetic evidence for shear along the Tethys. In: C. G. ADAMS and D. V. AGER (Editors), *Aspects of Tethyan Biogeography—Systematics Assoc. Publ.*, 7: 59–76.

KRYLATOV, S., 1963. Stratigraphie des terrains dinantiens de l'Argentella Tour Margine (NW de la Corse). *Bull. Soc. Géol. France*, 5: 312–317.

KRYLATOV, S. et MAMET, B., 1966. Données nouvelles sur les terrains paléozoiques de l'Argentella-Tour Margine. *Bull. Soc. Géol. France*, 8: 73–79.

LAPADU-HARGUES, P. et MAISONNEUVE, J., 1963. L'âge des schistes lustrés de la Corse. *Bull. Soc. Géol. France*, 5: 1012–1028.

MURAOUR, P., MARCHAND, J. P., DUCROT, J. et CECCALDI, X., 1966. Remarques sur la structure profonde du pré-continent de la région de Calvi. *Compt. Rend.*, 262: 17–19.

NAIRN, A. E. M. and WESTPHAL, M., 1967a. A second virtual pole from Corsica, the Ota gabbro-diorite. *Palaeogeography, Palaeoclimatol., Palaeoecol.*, 3: 277–286.

NAIRN, A. E. M. and WESTPHAL, M., 1967b. Sur l'aimantation de rhyolites permiennes en Corse. *Compt. Rend.*, 265D: 319–321.

PAYO, A., 1966. Crustal structure of the Mediterranean Sea by surface waves, 4. Group velocity. *Spanish Seismol. Surv., Sci. Rept., 15; and Bull. Seismol. Soc. Am.*, 52(1967): 151–172.

PETERSCHMITT, E., 1967. Propagation des ondes Lg à travers l'Afrique et la Méditerranée occidentale. *Ann. Inst. Globe Strasbourg*, 8(3): 85–87.

RECQ, M., 1967. Structure de la croute terrestre en Provence d'après les expériences du Revest et du Lac Nègre, *Compt. Rend.*, 264D: 1588–1591.

ROCHE, A., 1957. Sur l'aimantation des roches volcaniques de l'Esterel. *Compt. Rend.*, 244: 2952–2954.

RUTTEN, M. G., 1963. Palaeomagnetism and tethys. *Geol. Rundschau*, 53: 9–16.

RUTTEN, M. G., EVERDINGEN, R. O. and VAN ZIJDERVELD, J. D. A., 1957. Palaeomagnetism in the Permian of the Oslo graben (Norway) and of the Esterel (France). *Geol. Mijnbouw*, 19: 193–195.

STRANGWAY, D. W., MCMAHON, B. E. and HONEA, R. M., 1967. Stable magnetic remanence in antiferromagnetic goethite. *Science*, 158: 785–787.

THELLIER, E. and RIMBERT, F., 1955. Sur l'utilisation en paléomagnétisme de la desaimantation par champs alternatif. *Compt. Rend.*, 240: 1404–1406.

THELLIER, E. et THELLIER, O., 1959. Sur l'intensité du champ magnétique terrestre dans le passé historique et géologique. *Ann. Géophys.*, 15: 285–376.

VAN HILTEN, D., 1962. A deviating Permian pole from rocks in northern Italy. *Geophys. J.*, 6: 377–390.

VAN TELLINGEN, H. W., 1955. *Géologie et Pétrologie de la Région de Porto (Corso)*. Thesis, Univ. Amsterdam, 124 pp.